



Society of Petroleum Engineers

SPE-230217-MS

Enhancing Well Reliability and Productivity: Implementing the 7-Inch Annulus Casing Packer Tieback System for 9-5/8-Inch Casing Remedial and Gas Lifting Completion in the UAE

M. El-Mahdy, M. Hussain, and M. Youssef, ADNOC Onshore; A. Al Menhali, Abu Dhabi National Oil Company; P. L. Mwansa, B. Banihammad, M. Husien, A. Sinaga, and N. Upadhyay, ADNOC ONSHORE; A. Elbarag, M. Mohammed, and I. Saif, Weatherford

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230217-MS](https://doi.org/10.2118/230217-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The application of the 7-inch Annulus Casing Packer (ACP) Tieback system in ADNOC Onshore's 9-5/8-inch casing remedial and gas-lift completion programs delivers notable operational and economic benefits. Certified to ISO 14998 V0 standards, which qualifies packer seals for zero gas migration, the ACP offers a superior alternative to conventional tieback methods, consistently shortening project schedules by 5–7 days. This performance is driven by its integrated design features, including a hydraulically actuated Stage Tool for annular access, a V0-rated packer, and a Landing Collar with a non-shearable ball seat. The non-shearable ball seat design removes the need for temporary isolation means below the top of the liner (TOL), eliminating the risk of cement slumping or falling into the reservoir instead of filling the annulus between the 9-5/8-inch casing and 7-inch Tieback casing.

This approach focuses on deploying the ACP Tieback system in combination with optimized annular sealing strategies, supported by mechanical integrity simulations to assess long-term performance under pressure and temperature cycling. Tieback installation in remedial scenarios—particularly where gas-lift mandrels are incorporated—presents unique challenges, including the requirement for accurate assessment of outer casing condition, burst pressure capacity, and selection of an appropriate top of seal. The methodology also incorporates displacement sequencing and interface management to ensure clean, uncontaminated sealing surfaces, thereby enhancing reliability.

Field implementation in more than 90 ADNOC Onshore wells has demonstrated consistent time savings, improved well integrity, and the elimination of multiple ancillary tools or interventions. This advancement contributes to operational efficiency, cost reduction, and sustainability objectives by minimizing rig time and equipment usage. The paper presents this solution as a proven, field-ready enhancement to completion and remedial operations, offering engineers a practical pathway to achieve superior performance with reduced complexity.

Introduction

The oil and gas industry continues to evolve under the dual pressures of enhancing well integrity and maximizing efficiency in both new drilling and remedial operations. In mature assets, causing integrity problems—such as leaks, corrosion, and mechanical damage—remain a persistent challenge, especially when combined with complex well designs and the demand to extend life. Operators are tasked with balancing the need for reliable integrity solutions against the economic imperative to reduce operational costs and rig time, without compromising safety or production performance. In this environment, fit-for-purpose technologies that deliver durability, operational flexibility, and cost efficiency are key to sustaining output and optimizing recovery.

ADNOC Onshore's diverse well stock—from high-deviation MRC wells to shallow completions—requires a modular, single-trip tieback that addresses localized casing damage and integrates cleanly with gas-lift and existing hardware. Conventional tieback-to-surface practice, setting bridge plugs or placing acid-soluble plugs below the TOL, adding isolation steps, material handling, and trips, driving up complexity and rig time.

The 7-inch Annulus Casing Packer (ACP) Tieback system offers a streamlined, high-integrity alternative. By incorporating a non-shearable ball seat installed inside a Landing Collar, the system eliminates the need for downhole isolation plugs, preventing downward migration of cement into the reservoir and ensuring proper annular sealing between the 9-5/8-inch host casing and the 7-inch tieback string. The V0-rated Annulus Casing Packer, hydraulically activated to provide an additional barrier above the liner top packer and the Seal Assembly, further enhances wellbore integrity.

A hydraulically operated multi-stage tool allows controlled annular access for displacement and cement placement operations, while the Tieback Seal Assembly, equipped with metal-to-metal WZ seals, delivers a robust gas-tight barrier.

The system's modular design also incorporates casing joints to restore mechanical integrity in the damaged interval, as well as an optional orifice collar, integrated with a one-way valve, to regulate fluid flow during placement, reducing the risk of poor sealing distribution and maintaining operational control. Together, these features enable remediation of causing integrity issues with minimal changes to the existing completion and preserve flexibility for future interventions.

Field deployment across more than 90 wells in ADNOC Onshore operations has demonstrated consistent reductions in operational time, typically by 5–7 days, while improving annular sealing performance and eliminating the need for ancillary isolation tools. By reducing rig time, material usage, and operational complexity, the ACP Tieback system supports ADNOC's broader objectives for cost optimization, reliability, and sustainability.

This paper presents the design, field application, and performance results of the 7-inch ACP Tieback system in remedial and gas-lift completion scenarios. It outlines the operational methodology; highlights lessons learned from large-scale deployment and discusses future optimization opportunities to further enhance reliability and efficiency across varied well types and operating conditions.

Background / Problem Statement

Maintaining well integrity in mature oilfields remains one of the most pressing operational challenges for upstream operators. Over time, decades of production subjects cause aggressive downhole environments, where corrosive fluids, pressure cycling, and mechanical stresses contribute to progressive deterioration. This deterioration can manifest as leaks, loss of zonal isolation, and annular pressure buildup, each posing risks not only to production efficiency but also to long-term reservoir management. Left unaddressed, such issues can escalate to sustained causing pressure, compromised completions, or even well abandonment, resulting in significant economic losses.

A key operational priority during remedial tieback-to-surface installations is preventing sealing material, i.e. cement or other barrier mediums, from migrating downward into the reservoir or lower completion. If such migration occurs, it can lead to formation damage, plugging critical completion hardware. In remedial operations, two conventional approaches are widely used to create a barrier below the Top of Liner (TOL) before installing the tieback:

- **Bridge Plug**

This method involves running a scraper to clean the liner top, setting a bridge plug to isolate the open hole or lower completion, and then preparing the TOL for tieback installation. Once the tieback is run and the sealing operation is complete, the bridge plug's rubber components are drilled out, and the plug is pushed to bottom. Although widely adopted, this approach is operationally intensive. Depending on wellbore conditions and depth, the full sequence can take between 4 and 9 days, representing a significant time and cost investment. Additionally, the drill-out phase carries the risk of generating debris that can cause further complications.

- **Acid-Soluble Cement Plug**

In this method, an acid-soluble balance plug is placed using a cement stinger to protect the lower completion or reservoir. This process takes roughly two days under ideal conditions, but the risk of losses during placement can extend the timeline substantially. If losses occur, remedial steps may need to be repeated, including re-preparing the TOL, running the 7-inch tieback, and reinitiating the placement operation. Following completion, the acid-soluble plug and any associated rubber goods must be drilled out—a process that adds at least another day. Under less-than-ideal conditions, the total duration can range from 5 to 9 days.

While these methods are technically effective in certain applications, they share common disadvantages. Both introduce multiple operational stages, require additional downhole hardware, and extend the number of trips in and out of the hole. The result is not only higher rig time and cost but also greater exposure to operational hazards such as swabbing, debris-induced restrictions, and well control complications. Extended operations also increase the window for unplanned events, which can further disrupt timelines and budgets.

These limitations create a strong operational and economic case for an alternative remedial tieback technology that can:

- Deliver a reliable, high-integrity annular seal without the need for additional downhole isolation plugs.
- Significantly reduce rig time and streamline the operational sequence.
- Maintain adaptability for various completion designs, including gas-lift systems.
- Provide consistent sealing performance across diverse wellbore conditions.

The 7-inch Annulus Casing Packer (ACP) Tieback system was developed to meet these requirements. By integrating a hydraulically set V0-rated annulus packer, a non-shearable ball seat within the landing collar, and a hydraulically activated multi-stage tool for annular access, the system addresses the drawbacks of conventional methods while enhancing sealing reliability. Large-scale field deployment in ADNOC Onshore wells has shown its ability to restore casing integrity, reduce operational timelines by 5–7 days, and eliminate the need for additional downhole isolation measures, positioning it as a proven, field-ready solution for addressing casing integrity problems in both remedial and gas-lift completion applications.

System Design and Technology Description

The 7-inch Annulus Casing Packer (ACP) Tieback System was developed to address the limitations of conventional remedial tieback methods by eliminating the need for dedicated isolation runs prior to the

cemented tieback operation. The system integrates multiple high-performance components into a single operational run, delivering a reliable seal between the tieback string and the existing casing, while also providing a robust barrier against annular gas or fluid migration.

System Components

The ACP Tieback System consists of four key elements, each serving a distinct function to ensure mechanical integrity and operational efficiency:

1. ***V0-Rated Hydraulic Stage Tool***

Certified to ISO 14998 V0 standards, the hydraulic stage tool enables cement to be pumped through the tieback casing into the annulus, ensuring effective zonal isolation. The tool's pressure-activated mechanism mitigates Re-Stinging Risks Post-Cement Job by providing means of circulation through Stage Tool ports.

2. ***V0-Rated Annulus Casing Packer***

Installed above the seal stem and existing liner top packer, the annulus casing packer serves as an additional high-integrity barrier. The packer's longer elastomer element ensures optimal sealing even in irregular casing IDs or ovalized wellbores, making it particularly effective in mature or damaged wells.

3. ***Landing Collar with Non-Shearable Ball Seat***

Being equipped with a non-shearable ball seat, the landing collar eliminates the need for bridge plugs or acid-soluble cement plugs, streamlining the completion process and reducing material costs.

4. ***Tieback Seal Stem***

incorporated with 4–6 WZ seals certified to ISO 14998 V0, provides a high-integrity pressure barrier against the ID of the 7-inch top of liner. These elastomeric WZ seals are designed to maintain sealing efficiency under both high-pressure differential and temperature cycling, ensuring long-term zonal isolation.

5. ***Orifice Collar***

equipped with dual Sure-Seal 3 one-way valves. This component offers an additional layer of well-control protection by allowing fluid flow in one direction only, preventing backflow into the wellbore during operations. Its integration can be particularly advantageous in high-pressure or uncertain well conditions, adding safety redundancy without compromising the system's hydraulic efficiency.

Technology Features

The ACP Tieback System delivers a combination of high-performance sealing, operational flexibility, and cost efficiency:

- ***ISO 14998 V0 Qualification***

- ISO (International Organization for Standardization) is an independent, non-governmental body that develops consensus-based international standards.
- ISO 14998 is a petroleum & natural gas standard for validating downhole sealing equipment (e.g., packers/bridge plugs).
- V0 is the most stringent grade: no detectable gas leakage ("zero-bubble") is permitted while the tool is tested under differential pressure, axial loads, and temperature cycling.

The technology components were specifically re-engineered and qualified to V0 standards, ensuring full compliance with ADNOC's standards.

- ***High Internal and External Pressure Ratings***

Allows deployment in challenging conditions with significant pressure variations, without compromising barrier integrity.

- ***Operational Versatility***

The compact and robust design enables deployment in wells of varying deviation and geometry, maintaining performance across both vertical and highly deviated wellbores.

- ***PDC-Drillable Components***

The components are designed with PDC-drillable materials to significantly reduce post-installation drill-out time. Integrated anti-rotation features enhance tool stability during drilling, preventing unwanted movement and ensuring an efficient drill-out process. By eliminating opening cones and reducing internal drillable accessories to the bare minimum, the design enables a smoother and faster drill-out, minimizes debris, and further reduces operational downtime.

- ***Simplified Installation***

Single-trip deployment reduces rig time and operational risk. The integrated design eliminates the need for additional bridge plugs, cleanout runs, or dedicated isolation trips.

Operational Methodology

The deployment of the 7-inch ACP Tieback System is designed as a single-trip operation to maximize efficiency and minimize rig time while ensuring optimal sealing and zonal isolation. This methodology carefully sequences mechanical engagement, hydraulic setting, and cement placement to achieve reliable integrity without requiring additional isolation plugs or remedial runs. The process incorporates precise control over pressure application and fluid displacement to maintain operational safety and maximize cement quality.

The key procedural steps are as follows:

1. The ACP Tieback assembly is run in holes with the casing string to the Top of Liner (TOL). Precise stinging of the Tieback Seal Stem into the liner top polished bore receptacle ensures proper mechanical engagement and establishes an initial sealing interface critical for subsequent operations. The Seal Stem seals are then pressure-tested from the annulus to confirm integrity and ensure reliable isolation.
2. An activation ball is dropped from surface and allowed to free-fall through the tieback casing string until it lands on the non-shearable ball seat integrated within the landing collar. The successful seating of the ball is confirmed via a surface pressure response, serving as a key indicator for tool activation readiness.
3. Pressure is increased in a controlled, staged manner to hydraulically set the Annulus Casing Packer. Further pressure is applied to actuate the hydraulic Stage Tool, re-establishing circulation to the annulus to facilitate fluid pumping into the annulus.
4. Spacer fluid is pumped ahead of the cement slurry to effectively displace drilling fluids and prevent slurry contamination, preserving the integrity and placement quality of the cement. Cement is then pumped according to the well-specific program, ensuring full coverage of the annulus above the ACP and securing a durable zonal barrier.
5. Upon completion of cement pumping, the closing plug is released from surface to displace the slurry at a controlled rate into the annulus. The closing plug functions as a mechanical barrier between the cement and the displacement fluid, preventing contamination and ensuring slurry integrity. In addition, it provides the necessary mechanism to hydraulically shift the stage tool closing sleeve into its permanently closed & locked position.
6. Once the closing plug has landed on the closing seat of the stage tool, an additional 1,200 psi above the final displacement pressure is applied to hydraulically close the Stage Tool. This closure is confirmed

by a distinctive pressure & flow signature, verifying the tool's permanently locked position and sealing integrity.

Field Implementation

The 7-inch ACP Tieback System has been successfully deployed in more than 90 ADNOC Onshore wells, covering a wide range of applications including gas-lift completions and remedial casing tiebacks. This extensive track record across ADNOC's diverse fields underscores the system's reliability, versatility, and the operator's confidence in its performance. By enabling single-trip deployment, the system eliminates the need for bridge plugs or acid-soluble plugs, delivering rig-time savings of 5–7 days and significantly reducing operational costs.

The system has consistently demonstrated reliable performance under varied pressure, temperature, and wellbore deviation conditions. Post-cementing pressure tests confirmed effective annular isolation with no indication of fluid migration. The non-shearable ball seat and seal stem design further facilitated smooth drill-out operations with minimal debris, reducing downtime and enhancing overall operational efficiency.

A key operational insight involves the closing plug bumping procedure: due to the absence of a Shoe Track, an over-displacement of approximately 5-8 barrels is required to account for fluid compressibility and volume uncertainties. This practice ensures the closing plug properly bumps onto the closing sleeve, enabling secure hydraulic closure of the Stage Tool.

Beyond primary tieback remedial applications, the ACP system has also proven particularly valuable in workover operations where the Lower Completion remains in place below the cemented liner. In such cases, the system eliminates the risk of debris migrating into the lower completion after drilling out the acid-soluble plug, thereby preventing damage to sensitive downhole equipment and avoiding costly remediation. This added protection further enhances the value proposition of the ACP system by safeguarding well integrity during various remedial interventions.

Collectively, these field results validate the ACP Tieback System as a proven, efficient, and versatile solution for strengthening well integrity and improving operational efficiency across ADNOC Onshore's broad portfolio of wells.

Discussion and Conclusions

The field deployment of the 7-inch ACP Tieback System across ADNOC Onshore wells provides a strong basis for assessing its operational and economic impact. Beyond the straightforward rig-time savings and elimination of ancillary isolation runs, the system's consistent performance under varied conditions demonstrates that it is not simply an incremental improvement but a step-change in remedial tieback practices. The integration of ISO 14998 V0-qualified sealing elements, combined with a non-shearable ball seat and annulus casing packer, ensures that high-integrity zonal isolation can be achieved with less operational complexity and reduced exposure to risks traditionally associated with bridge plugs and acid-soluble plugs.

One of the most significant findings from large-scale implementation is the robustness of the system across diverse well environments. Deployments were not limited to straightforward vertical wells but included highly deviated geometries, challenging pressure regimes, and fields with a history of causing integrity issues. The absence of gas migration during post-installation pressure tests consistently validated the reliability of the sealing system and highlighted the effectiveness of precise displacement management.

The technology has proven to be most robust in workover scenarios where the Lower Completion remains in place. In such cases, the ACP system provides a safeguard against debris ingress, mitigating a long-standing risk that can compromise expensive downhole equipment and lead to costly interventions. The ability to preserve the integrity of the Lower Completion during remedial tieback operations is a differentiator that extends the value of the system beyond conventional cementing or casing repair functions.

This feature is particularly relevant for operators managing mature fields, where remedial workovers are both frequent and technically challenging.

From an operational efficiency perspective, the reduction of 5–7 rig days per installation translates not only into direct cost savings but also into a broader contribution to ADNOC's sustainability objectives. By minimizing equipment requirements, limiting unnecessary trips in and out of the hole, and reducing exposure to unplanned events, the system reduces energy consumption and lowers the carbon footprint of remedial operations. These factors align with industry-wide efforts to balance production optimization with environmental responsibility.

The results also indicate opportunities for further optimization. While the system has proven reliable across a wide range of conditions, continued refinement of displacement strategies, pressure ramp-up profiles, and contingency planning could enhance operational resilience even further. Data collected from ADNOC's large deployment base creates a valuable foundation for predictive modeling and digital integration, enabling engineers to anticipate well-specific behaviors and tailor operational parameters accordingly.

In conclusion, the ACP Tieback System has established itself as a reliable, field-ready technology that addresses the limitations of conventional remedial tieback approaches. Its proven ability to deliver high-integrity annular sealing, streamline operations, and protect lower completions has made it a cornerstone of ADNOC Onshore's remedial and gas-lift completion programs. The combination of technical robustness, operational flexibility, and measurable economic benefits positions the system as a best practice for casing integrity restoration in mature assets. Future work should continue to build on these successes by integrating digital monitoring and expanding deployment in other regional contexts, ensuring that the technology's value is maximized both within ADNOC Onshore and across the wider industry.